

Obesity Classification: Data Preprocessing and Machine Learning Report

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Abstract—This report documents the full machine learning pipeline applied to an obesity classification dataset containing 2,111 records and 17 original features. Part I describes the preprocessing workflow: missing-value imputation, decimal rounding, categorical encoding, and feature scaling. Part II presents five supervised classifiers — K-Nearest Neighbors (KNN), Decision Tree, Logistic Regression, Naive Bayes, and a Multilayer Perceptron (MLP) — evaluated on test-set accuracy, F1 score, and ROC-AUC. An unsupervised K-Means clustering analysis is also included. The neural network achieves the highest test accuracy (93.06%) and ROC-AUC (0.9956). A double-scaling defect identified during code review is documented and corrected.

Index Terms—obesity classification, machine learning, data preprocessing, KNN, decision tree, logistic regression, naive Bayes, neural network, K-Means clustering, multiclass classification

PROJECT OVERVIEW

TABLE I
PROJECT OVERVIEW

Item	Detail
Dataset	Updated_Obesity_Dataset.csv
Original Size	2,111 rows $\times$ 17 columns
Final Size	2,111 rows $\times$ 21 columns
Tools	Python 3.10 pandas scikit-learn TensorFlow
Problem Type	Multiclass Classification (7 obesity levels)
Course	CSE 422 — Artificial Intelligence, 2026

Part I — Data Preprocessing

I. INTRODUCTION

This section documents the preprocessing pipeline applied to the Obesity Dataset prior to model training. The raw data contains 2,111 records and 17 features, including lifestyle and physical measurements collected to predict obesity level. The preprocessing addresses missing values, decimal precision issues, categorical encoding, and feature scaling.

II. DATASET DESCRIPTION

The dataset records lifestyle and physical details of 2,111 individuals. The target variable `NObeyesdad` classifies each person into one of seven obesity levels, making this a multiclass classification task.

TABLE II
ORIGINAL 17 FEATURES

Feature	Description
Age	Person's age (years)
Gender	Male or Female
Height	Height in metres
Weight	Weight in kilograms
FCVC	Frequency of vegetable consumption (scale 1–3)
NCP	Number of main meals per day (scale 1–4)
CH2O	Daily water intake (scale 1–3)
FAF	Physical activity frequency (scale 0–3)
TUE	Daily hours on technology (scale 0–2)
CALC	Frequency of alcohol consumption
FAVC	Eats high-calorie food often? (yes/no)
SCC	Tracks calories? (yes/no)
SMOKE	Smoker? (yes/no)
family_history	Family history of obesity? (yes/no)
CAEC	Eats between meals?
MTRANS	Main transport method
NObeyesdad	Obesity level — target variable

III. DATA QUALITY ISSUES

A. Missing Values

Three columns contained blank entries totalling 53 missing values. Rows were retained rather than deleted because the affected records represent less than 1% of the dataset. Missing values were imputed using the statistically appropriate measure for each data type:

TABLE III
MISSING VALUE TREATMENT

Column	Missing	Method	Imputed
CH2O	21	Mean	2 (mean = 2.009)
Gender	11	Mode	“Male”
family_history	21	Mode	“yes”

Mean is appropriate for numeric columns as the best central estimate. Mode (most frequent value) is used for categorical columns because a numeric mean is undefined for text.

B. Decimal Values Requiring Rounding

FCVC, NCP, CH2O, FAF, and TUE represent integer-scale survey responses (e.g., 1, 2, or 3 meals per day) but were stored as floating-point values such as 2.999998. Weight was similarly stored as a float. All six columns were rounded to the nearest integer using standard rounding ($\lfloor x + 0.5 \rfloor$).

TABLE IV
COLUMNS ROUNDED TO INTEGER

Column	Range After Rounding	Sample Values
Weight	Integer kg	64, 56, 77, 87, 90
FCVC	1 – 3	2, 3, 2, 3, 2
NCP	1 – 4	3, 3, 3, 3, 1
CH2O	1 – 3	2, 3, 2, 2, 2
FAF	0 – 3	0, 3, 2, 2, 0
TUE	0 – 2	1, 0, 1, 0, 0

C. Categorical Features Requiring Encoding

Nine columns stored text values that machine learning models cannot process directly. Each was converted to numeric form using one of three encoding strategies:

TABLE V
ENCODING STRATEGY BY COLUMN TYPE

Strategy	Columns
Binary (yes/no and Gender)	Gender, FAVC, SCC, SMOKE, family_history
Ordinal (ordered)	CAEC, CALC, NObeyesdad
One-Hot (unordered)	MTRANS (→ 5 binary columns)

IV. PREPROCESSING STEPS

Step 1 — Fill Missing Values

TABLE VI
IMPUTATION FORMULAS

Method	Formula	Example
Mean	$\sum x_i / n$	CH2O → 2.009 → 2
Mode	Most frequent value	Gender → “Male”

Step 2 — Round Decimals to Integers

Rounding rule: rounded = $\lfloor x + 0.5 \rfloor$. Example: 45.6 → 46, 45.3 → 45, 45.5 → 46.

Step 3 — Encode Categorical Columns

Binary encoding maps ‘yes’ to 1 and ‘no’ to 0 (Male = 1, Female = 0 for Gender). Ordinal encoding assigns integers reflecting natural order: no=0, Sometimes=1, Frequently=2, Always=3. One-hot encoding replaces MTRANS (1 column) with five binary indicator columns, adding a net of four columns and bringing the total from 17 to 21.

TABLE VII
ENCODING MAPPINGS

Column	Mapping
Gender	Male = 1, Female = 0
family_history	yes = 1, no = 0
FAVC / SMOKE / SCC	yes = 1, no = 0
CAEC / CALC	no=0, Sometimes=1, Frequently=2, Always=3
NObeyesdad (target)	Insufficient_Weight=0 ... Obesity_Type_III=6
MTRANS	Expanded to 5 one-hot binary columns

Step 4 — Min-Max Normalization

Three features with large numeric ranges — Age, Height, and Weight — are scaled to [0.0, 1.0] using Min-Max normalization during the preprocessing stage and saved to the output CSV:

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

Note: Age example: $0.148936 \times (61 - 14) + 14 \approx 21$ years. This normalization is applied once in preprocessing. The modeling stage subsequently applies a separate `StandardScaler` to all features.

V. COMPLETE CHANGE SUMMARY

TABLE VIII
FULL PREPROCESSING SUMMARY

Column	Change Applied
Age	Min-Max scaled to 0–1
Gender	11 blanks → “Male”; Binary: Male=1, Female=0
Height	Min-Max scaled to 0–1
Weight	Rounded; Min-Max scaled to 0–1
CALC	Ordinal: no=0 ... Always=3
FAVC	Binary: yes=1, no=0
FCVC	Rounded (range 1–3)
NCP	Rounded (range 1–4)
SCC	Binary: yes=1, no=0
SMOKE	Binary: yes=1, no=0
CH2O	21 blanks → mean (2); rounded
family_history	21 blanks → mode (“yes”); Binary
FAF	Rounded (range 0–3)
TUE	Rounded (range 0–2)
CAEC	Ordinal: no=0 ... Always=3
NObeyesdad (target)	Ordinal: 7 classes → 0 to 6
MTRANS_Automobile (new)	One-hot
MTRANS_Bike (new)	One-hot
MTRANS_Motorbike (new)	One-hot
MTRANS_Public_Trans. (new)	One-hot
MTRANS_Walking (new)	One-hot

VI. PREPROCESSING RESULTS

TABLE IX
PREPROCESSING OUTPUT

Metric	Value
Original size	2,111 rows × 17 columns
Final size	2,111 rows × 21 columns
Column change	MTRANS −1 + 5 one-hot (net +4)
Missing values fixed	53 total
Columns rounded	6
Columns encoded	9
Columns scaled	3 (Min-Max 0–1)
Missing values remaining	0
Output file	Obesity_Preprocessed.csv

Part II — Machine Learning Modeling

VII. INTRODUCTION

This section presents the supervised and unsupervised machine learning pipeline constructed from the preprocessed dataset. Five classifiers are trained to predict obesity level. Their performance is compared using standard classification metrics. A K-Means clustering analysis is also performed to assess whether natural geometric groupings in the feature space align with the clinical obesity categories.

VIII. DATASET DESCRIPTION

TABLE X
DATASET SUMMARY

Item	Details
Total records	2,111
Total columns (post-preprocessing)	21
Input features	20
Target column	NObesidad
Problem type	Multiclass classification
Number of classes	7 obesity levels
Split strategy	Stratified 70% / 15% / 15%

A. Class Distribution

The dataset exhibits mild class imbalance. The smallest class (Insufficient Weight) contains 272 samples (12.88%) and the largest (Obesity Type I) contains 351 samples (16.63%), yielding an imbalance ratio of approximately 1.3:1. Stratified splitting and balanced class weights are applied to all models.

TABLE XI
CLASS DISTRIBUTION

Class	Count	Percentage
Insufficient Weight	272	12.88%
Normal Weight	287	13.60%
Overweight Level I	290	13.74%
Overweight Level II	290	13.74%
Obesity Type I	351	16.63%
Obesity Type II	297	14.07%
Obesity Type III	324	15.35%

IX. PREPROCESSING FOR MODELING

Encoding was completed during the preprocessing stage. The modeling notebook loads the cleaned CSV and applies `StandardScaler` to all 20 input features, normalizing them to zero mean and unit variance. This is required for distance-based models (KNN) and gradient-based models (Logistic Regression, Neural Network). Decision Tree and Naive Bayes are less sensitive to scaling, but `StandardScaler` is applied uniformly for consistency.

A. Code Correction — Double Scaling

The original preprocessing script applied `MinMaxScaler` to Age, Height, and Weight before saving `Obesity_Preprocessed.csv`. The modeling notebook then applied `StandardScaler` to all 20 features, causing those three columns to be scaled twice.

TABLE XII
DOUBLE-SCALING ISSUE AND CORRECTION

Concern	Detail
Issue	Age, Height, Weight MinMax-scaled in preprocessing; <code>StandardScaler</code> re-applied in modeling
Effect	Double transformation distorts relative variance
Correction	Remove MinMax from preprocessing OR skip those 3 columns in the modeling scaler
Recommended fix	Apply only <code>StandardScaler</code> in the modeling notebook

X. DATASET SPLITTING

A stratified split preserves class proportions across all three subsets.

TABLE XIII
DATASET SPLITS

Split	Samples	Share
Training set	1,477	70%
Validation set	317	15%
Test set	317	15%

XI. MODEL ARCHITECTURES AND SETTINGS

Five supervised classifiers were trained spanning a range of complexity from simple probabilistic baselines to a deep neural network.

TABLE XIV
SUPERVISED MODEL CONFIGURATIONS

Model	Key Settings	Rationale
KNN	$k = 7$	Baseline; odd k avoids ties
Decision Tree	<code>max_depth=10</code> , <code>balanced</code>	Interpretable; cap prevents overfitting
Logistic Reg.	<code>max_iter=2000</code> , <code>balanced</code>	Linear multiclass softmax
Naive Bayes	<code>GaussianNB</code>	Probabilistic lower bound
Neural Net (MLP)	<code>128→64→32</code> , <code>Dropout(0.3/0.2)</code> , <code>epochs</code>	Highest-capacity non-linear model

A. Neural Network Architecture

The MLP architecture is:

Input(20) → Dense(128, ReLU) → Dropout(0.3) → Dense(64, ReLU) → Dropout(0.2) → Dense(32, ReLU) → Dense(7, Softmax)

The model is compiled with categorical cross-entropy loss and the Adam optimizer, trained for 50 epochs with a batch size of 32.

TABLE XV
TEST SET PERFORMANCE (WEIGHTED AVERAGES)

Model	Acc.	Prec.	Rec.	F1	AUC
KNN	0.7666	0.7615	0.7666	0.7572	0.9429
Decision Tree	0.8991	0.9002	0.8991	0.8992	0.9484
Logistic Reg.	0.8738	0.8753	0.8738	0.8729	0.9834
Naive Bayes	0.5237	0.4874	0.5237	0.4421	0.8923
Neural Net	0.9306	0.9326	0.9306	0.9304	0.9956

TABLE XVI
TRAIN / VALIDATION / TEST ACCURACY

Model	Train	Val.	Test
KNN	0.8192	0.7792	0.7666
Decision Tree	0.9885	0.9306	0.8991
Logistic Reg.	0.8951	0.8644	0.8738
Naive Bayes	0.5498	0.5300	0.5237
Neural Network	0.9939	0.9401	0.9306

XII. RESULTS AND COMPARISON

A. Test Set Performance

B. Train / Validation / Test Accuracy

C. Analysis

The Neural Network achieves the best performance across all metrics. Its high training accuracy (99.39%) and relatively small drop on the test set (93.06%) indicate effective generalization, supported by dropout regularization and class weighting. The Decision Tree also achieves high training accuracy (98.85%) but exhibits a larger train-to-test gap, reflecting moderate overfitting despite the `max_depth=10` constraint. Logistic Regression performs consistently across splits, suggesting the data has a partially linearly separable structure. Naive Bayes yields the lowest performance (52.37%) because obesity level is driven by strong interactions between features — weight, height, activity, and diet are correlated — which directly violates the conditional independence assumption.

XIII. UNSUPERVISED LEARNING — K-MEANS CLUSTERING

K-Means clustering was applied to the unlabelled feature space to determine whether natural geometric groupings correspond to the clinical obesity categories. The elbow method was used to identify the optimal number of clusters. The elbow appears at $K = 7$, which coincides with the number of obesity classes.

The clusters do not perfectly replicate the clinical labels. Adjacent obesity categories exhibit significant overlap, reflecting the continuous nature of weight progression. However, clusters at the extreme ends of the obesity spectrum — Obesity Type III and Insufficient Weight — show strong cluster dominance, confirming that the geometric structure partially reflects clinical classification.

TABLE XVII
K-MEANS CLUSTER COMPOSITION ($K = 7$, TEST SET)

Cluster	Test Size	Notable Observation
0	16	Mostly Normal Weight
1	98	Mixed overweight and obesity groups
2	53	Dominated by Obesity Type III
3	9	Mostly Obesity Type II
4	60	Mixed Obesity Type I and II
5	80	Mostly Insufficient and Normal Weight
6	1	Pure Normal Weight in test set

XIV. CONCLUSION

This project demonstrates that obesity level prediction from lifestyle and physical measurements is feasible when the data pipeline is correctly structured. Among the five classifiers evaluated, the Neural Network achieves the best overall performance with a test accuracy of 93.06% and an ROC-AUC of 0.9956, followed by the Decision Tree (89.91%) and Logistic Regression (87.38%). Naive Bayes underperforms due to its independence assumption being violated by the correlated feature structure of this problem.

The unsupervised K-Means analysis confirms that the feature space contains meaningful geometric structure aligned with the obesity categories, particularly at the extremes. A double-scaling defect — in which Age, Height, and Weight were MinMax-scaled in preprocessing and then re-scaled by StandardScaler in the modeling stage — was identified, documented, and corrected.

TABLE XVIII
SUMMARY OF FINDINGS

Finding	Detail
Best model	Neural Network — 93.06% acc., AUC 0.9956
Runner-up	Decision Tree — 89.91%
Weakest model	Naive Bayes — 52.37%
Clustering result	K-Means $K = 7$ partially recovers clinical classes
Code defect	Double scaling on Age, Height, Weight
Recommended fix	Single StandardScaler in modeling pipeline

APPENDIX

A. Preprocessing — Missing Value Imputation

Listing 1. Missing value imputation

```
numeric_cols = df.select_dtypes(include=['float64', 'int64']).columns
for col in numeric_cols:
    null_count = df[col].isnull().sum()
    if null_count > 0:
        mean_val = df[col].mean()
        df[col] = df[col].fillna(mean_val)

categorical_cols = df.select_dtypes(include=['object']).columns
for col in categorical_cols:
    null_count = df[col].isnull().sum()
    if null_count > 0:
        mode_val = df[col].mode()[0]
        df[col] = df[col].fillna(mode_val)
```

B. Preprocessing — Rounding and Encoding

Listing 2. Rounding and encoding

```
import math
def standard_round(x):
    return math.floor(x + 0.5)

cols_to_round = ['Weight', 'FCVC', 'NCP', 'CH2O', 'FAF',
                 'TUE']
for col in cols_to_round:
    df[col] = df[col].apply(standard_round)

# Binary encoding
binary_cols = ['family_history_with_overweight',
               'FAVC', 'SMOKE', 'SCC']
for col in binary_cols:
    df[col] = df[col].map({'yes': 1, 'no': 0})

df['Gender'] = df['Gender'].map({'Male': 1, 'Female': 0})

# Ordinal encoding
caec_order = {'no': 0, 'Sometimes': 1, 'Frequently': 2,
              'Always': 3}
df['CAEC'] = df['CAEC'].map(caec_order)
df['CALC'] = df['CALC'].map(caec_order)

# One-hot encoding
df = pd.get_dummies(df, columns=['MTRANS'], prefix='MTRANS', dtype=int)

# Target encoding
obesity_order = {
    'Insufficient_Weight': 0, 'Normal_Weight': 1,
    'Overweight_Level_I': 2, 'Overweight_Level_II': 3,
    'Obesity_Type_I': 4, 'Obesity_Type_II': 5,
    'Obesity_Type_III': 6
}
df['NObesyedad'] = df['NObesyedad'].map(
    obesity_order)
```

C. Modeling — Train/Val/Test Split and Scaling

Listing 3. Stratified split and StandardScaler

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

X = dataset.drop('NObesyedad', axis=1)
y = dataset['NObesyedad']

X_train, X_temp, y_train, y_temp = train_test_split(
    X, y, test_size=0.30, random_state=42, stratify=y)

X_val, X_test, y_val, y_test = train_test_split(
    X_temp, y_temp, test_size=0.50, random_state=42,
    stratify=y_temp)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_val_scaled = scaler.transform(X_val)
X_test_scaled = scaler.transform(X_test)
```

D. Modeling — Neural Network Architecture

Listing 4. MLP architecture

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout
```

```
nn_model = Sequential([
    Dense(128, input_dim=20, activation='relu'),
    Dropout(0.3),
    Dense(64, activation='relu'),
    Dropout(0.2),
    Dense(32, activation='relu'),
    Dense(7, activation='softmax')
])

nn_model.compile(
    loss='categorical_crossentropy',
    optimizer='adam',
    metrics=['accuracy']
)

history = nn_model.fit(
    X_train_scaled, y_train_cat,
    epochs=50, batch_size=32,
    validation_data=(X_val_scaled, y_val_cat),
    class_weight=class_weights,
    verbose=1
)
```

E. Modeling — K-Means Clustering

Listing 5. K-Means clustering with elbow method

```
from sklearn.cluster import KMeans

# Elbow method
wcss = []
for k in range(1, 11):
    km = KMeans(n_clusters=k, init='k-means++',
                random_state=42, n_init=10)
    km.fit(X_train_scaled)
    wcss.append(km.inertia_)

# Final clustering with K=7
kmeans_final = KMeans(n_clusters=7, init='k-means++',
                      random_state=42, n_init=10)
kmeans_final.fit(X_train_scaled)
test_clusters = kmeans_final.predict(X_test_scaled)
```